

Ruiyi Zhang

PhD Student in Applied Mathematics

AI-assisted mathematical research

Seeking an LLM Data Internship (Mathematics) in Beijing

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Education

Beihang University, PhD in Applied Mathematics

September 2024 to present

Expected graduation: 2029. Advisors: Changxing Miao and Xiaoxin Zheng.

Research: nonlinear PDEs, chemotaxis, aggregation-diffusion, degenerate diffusion, and energy estimates.

Beihang University, BSc in Mathematics and Applied Mathematics

2020 to 2024

Hua Loo-Keng Experimental Class for Mathematics. September 2020 to July 2024. GPA: 3.6/4.0.

Papers and Research

01 Small-Diffusivity Concentration for Fractional Aggregation with Cusp Attraction

Submitted to Mathematical Methods in the Applied Sciences (Wiley)

Ruiyi Zhang, Xiaoxin Zheng

- Study mass concentration in the competition between weak diffusion and cusp attraction, turning a scale balance into a quantitative mass bound at the intrinsic scale.
- Combine tent moments, nonlocal boundary decomposition, radial geometry, and Lorentz-space estimates to obtain a diffusivity-uniform time-integrated mass bound under radial-data and explicit time and scale conditions.

02 Time-Uniform Boundedness and Large-Time Behavior for a Planar Doubly Degenerate Nutrient-Taxis System

Manuscript in preparation

Ruiyi Zhang, Xiaoxin Zheng

- Study whole-plane weak solutions and time-uniform control when diffusion degenerates with both population density and nutrient concentration.
- Combine a compensated Lyapunov functional, radius-uniform endpoint inequalities, moment iteration, and compactness to construct globally bounded weak solutions and study regularity and long-time behavior.

03 Proof or Obstruction: Verifier-Guided Diagnosis of PDE Energy Estimates

Research manuscript v2

Ruiyi Zhang

- Translate parameter and budget searches within fixed templates into convex optimization, and use category theory to organize the composition and reuse of estimates.
- Prototype: eight source-backed routes and eight controlled perturbations, a 24-task model-feedback experiment, and deterministic baselines; explore model proposals combined with external verification.

Skills and Interests

Mathematics and AI: ChatGPT and Codex for derivations, energy-functional exploration, and research organization; interests in open analysis/PDE problems, reasoning evaluation, and proof checking.

Tools: Python (experience with NumPy, Pandas, and SciPy), basic C++, Linux terminal, and LaTeX.

English: CET-6; reading mathematical literature and writing research manuscripts in English.

Teaching and Honors

Teaching assistant, Beihang: PDEs (2025, 2026); Mathematical Analysis for Engineering (2025); Fundamentals of Computing (2021 to 2023).

Competitions: National College Mathematics Competition, Mathematics A, Second Prize (2023); Beijing Mathematics Competition, Category A, First Prize (2022); Mathematical Modeling Competition, Beijing Division, Category A, Second Prize (2022).

Academic program: First Prize, introductory course, Nonlinear PDE Summer School, Capital Normal University (2025).

Two PDE Papers: Motivation and Methods

My research concerns the competition between diffusion and aggregation, and how to obtain effective control when diffusion degenerates. Both papers below are joint work with my advisor, Xiaoxin Zheng; the author order is Ruiyi Zhang, Xiaoxin Zheng.

From Scale Balance to Quantitative Concentration

Small-Diffusivity Concentration for Fractional Aggregation with Cusp Attraction

Submitted to Mathematical Methods in the Applied Sciences (Wiley)

Motivation Attraction drives aggregation, while diffusion opposes concentration. Classical diffusion problems offer useful starting points, but fractional diffusion is nonlocal: even when a test function is supported in a small ball, mass outside the ball enters the calculation. How can the balance between attraction and diffusion lead to a rigorous concentration estimate?

We consider

$$\partial_t u + \varepsilon(-\Delta)^{\gamma/2} u + \nabla \cdot (u(\nabla K * u)) = 0, \quad K(x) = -|x|, \quad 1 < \gamma < 2.$$

Approach A tent-shaped truncated moment connects the test function to local mass. We decompose the fractional operator's cross-boundary contributions, use radial geometry to obtain a Riccati-type inequality, and apply Lorentz-space estimates to convert weighted quantities into small-ball mass.

The scale balance gives $R_\varepsilon = \varepsilon^{1/(\gamma-1)}$. For radial, nonnegative, integrable and bounded initial data, and under explicit conditions on T and λ , the paper obtains

$$\int_0^T \int_{B_{\lambda R_\varepsilon}} u^\varepsilon(x, t) dx dt \geq C_* > 0,$$

with C_* uniform for sufficiently small ε . This turns the formal concentration scale into a quantitative lower bound.

Mathematical work Choosing equation-adapted observables, handling integrability and signs for a nonlocal operator, and connecting geometry, differential inequalities, and function-space estimates into a proof route.

Energy Estimates under Double Degeneracy

Time-Uniform Boundedness and Large-Time Behavior for a Planar Doubly Degenerate Nutrient-Taxis System

Manuscript in preparation

Motivation Population diffusion depends on both density and nutrient concentration. As either approaches zero, diffusion weakens, while the population continues to consume nutrients. Understanding whole-plane evolution calls for an energy structure that handles this degeneracy and yields control independent of time and approximation-domain size.

The system is

$$u_t = \nabla \cdot (uv \nabla u) - \nabla \cdot (u^2 v \nabla v) + uv, \quad v_t = \Delta v - uv.$$

Approach We combine flux-adapted chemical entropy, a fourth-order logarithmic nutrient-gradient quantity, and nutrient-mass compensation in a Lyapunov functional. Radius-uniform endpoint inequalities yield a $p \mapsto 3p - 1$ moment iteration. Shifting by the fixed point $1/2$ organizes the recursion and controls population amplitude. Domain exhaustion and Aubin-Lions compactness then handle nonlinear flux limits.

Progress Under the stated initial-data and weighted gradient-energy assumptions, the manuscript constructs globally bounded weak solutions and studies positive-time regularity, nutrient decay, and long-time population convergence.

Mathematical work Designing energy functionals, absorbing cross terms into dissipation, organizing exponent iterations, and carrying a priori estimates through to weak-solution construction and qualitative analysis.

PDE Energy Search: Structure and Exploration

Proof or Obstruction: Verifier-Guided Diagnosis of PDE Energy Estimates

Ruiyi Zhang | Research manuscript v2 | July 2026

Motivation: A Different Way to Search for Exponents

PDE energy methods often produce $E'(t) + D(t) \leq B(t)$. The next step is to combine inequalities so that the troublesome term B fits within the available dissipation budget. In practice, I spend substantial time trying interpolation exponents, choosing Young parameters, checking whether conditions across steps hold together, and looking for better intermediate quantities.

This led me to ask: **can this estimate search be expressed through optimization and proof composition, so that successful routes and failures become easier to check?** Within suitable finite templates, exponent and parameter compatibility can be written as linear or convex constraints. Category theory provides a language for composing estimate steps, using them in parallel, and reusing their outputs.

What These Two Perspectives Contribute

Convex optimization and duality With rules and templates fixed, I express shared parameter conditions as a feasible set and dissipation costs as objectives or constraints. In linear cases, linear programming finds feasible parameters, while Farkas certificates or optimization duals explain infeasibility and insufficient budgets. The success or failure of an estimate route then has evidence that can be revisited.

Category theory and proof composition I treat estimates carrying domains, function spaces, and boundary conditions as objects, and inequality instances as composable inference steps. This separates abstract exponent calculations, concrete realization of analytic conditions, and global compatibility across local choices.

Diagnosis The framework distinguishes missing conditions, parameter conflicts, insufficient dissipation budgets, and non-reachability within a rule library. Each diagnosis suggests a different next step, such as adding an assumption, adjusting parameters, introducing a cross-energy term, or searching for a new inequality.

Prototype and Experiments

- Eight estimate routes extracted from eight published PDE papers, with eight single-factor perturbations, test route replay and failure diagnosis.
- A 24-task synthetic experiment compares diagnostic feedback. Four source-backed tasks are also tested under two prompt conditions, giving eight model calls.
- Both prompt conditions and deterministic baselines reconstruct all four tasks. Exact solving within fixed templates is an effective baseline, and points towards the next research focus: new representations, energy terms, or lemmas.

How Large Language Models Can Participate

Models can attempt complete derivations directly, generate many candidates, search broadly, and repeatedly revise their approaches. Convex optimization and category theory help organize, verify, and explain those candidates. Models remain free to explore without following a fixed formal system.

I hope to combine broad model exploration of energy functionals, variable transformations, and estimate routes with mathematical analysis and executable checks. During an internship, I plan to select problems relevant to my analysis and PDE background from collections such as ICM Conjectures, check the literature, and break them into intermediate propositions. I want to make meaningful mathematical progress while understanding where models can help.